CE C30 / ME C85: Introduction to Solid Mechanics

Required Textbooks

[GHSWR] *Engineering Mechanics 1: Statics* by Gross, Hauger, Schröder, Wall, and Rajapakse: Springer, 2nd edition (2013), ISBN: 9783642303180.[1]

[G] *Engineering Mechanics of Deformable Solids* by Govindjee: Oxford University Press (2013), ISBN: 9780199651641.

Prerequisites

Mathematics 53 and 54 (may be taken concurrently); Physics 7A

Reading

Assignments are required. Reading assignments may contain material not covered in lecture; you are responsible for this additional material.

CYU quizzes

After every lecture there will be a multiple choice check your understanding quiz to be completed on bCourses. These quizzes will often take no more than 10-20 minutes of your time and should be easily finished based on the associated lecture. Occasionally, they may take a bit more time. Quizzes are open from the end of lecture until 11:59pm of the same day. This gives you a 14-hour time window to complete a 10-20 minute assignment. No extensions will be granted for CYU quizzes.

Homework

Assignments are due every Friday at 5:59pm. They will always be comprised of the problems associated with the lectures of the prior week. For example, the homework for lectures 1 and 2 will be due September 4, the homework for lectures 3, 4, and 5 will be due on September 11, etc. Homework is to be uploaded to Gradescope with properly marked pages. Late

[1] [GHSWR] has an optional solution manual for its end of chapter problems which is available electronically from Amazon: *Engineering Mechanics 1, Supplementary Problems: Statics*

submissions accepted up to 24 hours late in case of emergencies. **All submissions must be hand-written.**[2]

Midterm Exams

Midterm 1 will be on September 21 and cover Lectures 1–10. Midterm 2 will be on November 2 and cover Lectures 11–26. Midterm exams are closed book, closed notes, but you may bring a single page of self-prepared **hand-written** notes to the exam. You may write on both sides of the note sheet which may not be larger than 8 1/2 x 11 inches in size (or optionally A4 size).[3]

Final Exam

The final examination is closed book, closed notes, but you may bring 3 sheets of self-prepared **hand-written** notes to the exam. You may write on both sides of the sheets which may not be larger than 8 1/2 x 11 inches in size (or optionally A4 size). The final exam is comprehensive and covers the entire course; it will take place December 17 from 7-10pm.[4]

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[2] All DSP accommodations for homework will automatically be entered in Gradescope. If there is a problem, please contact the head GSI.

[3] All DSP accommodation exams must be taken at the DSP testing center. You may start your midterm exams no earlier than 8am on the exam date and must finish before the testing center closes on that date.

[4] All DSP accommodation exams must be taken at the DSP testing center. You may start your final exam no earlier than 7pm December 17 and must finish before the testing center closes on Friday December 18.

Course Grade

Check your understanding questions 5%, Homework 20%, Midterm 1 15%, Midterm 2 15%, Final exam 45%.

Grading methodology

Every question (or sub-part for longer questions) will be graded on a scale of 0–4 – so on a GPA scale. This implies, if you receive a 3 on a question you can interpret that as a B grade. This allows you to easily keep track of where you stand in the course. The rubric for the scoring is as follow:

- **4** You have shown an excellent understanding of how to solve the problem and have gotten it correct.
- **3** You have shown a good understanding of how to solve the problem but have at least one conceptual error/omission. In large part you have solved the problem.
- **2** You have shown some understanding of the question but have at least two conceptual errors/omissions.
- **1** You have shown that you know what is being asked and perhaps know some of the applicable principles that are relevant to the question.
- **0** You have not attempted the problem or the work shown is largely irrelevant.

The only exception to this rule is that each CYU questions, which are multiple choice, and graded as either 0 or 2.
Note that illegible, sloppy, un-professional, etc. work will be scored 0.

Limited Collaboration

Limited collaboration is permitted on homework assignments and CYU quizzes. You may freely discuss the questions with each other, it is fine to explain how to solve an assignment, but you may not show your written work to others or directly tell someone which option to pick on a CYU quiz.

The only exception to these rules are for exams, which are exclusively individual assessments. Misconduct on examinations will be reported to the Center for Student Conduct and result in an automatic failing grade for the course.

Course Outline

Lecture	Module	Topic	Reading	HW
1: 8/26	Statics	Review of static equilibrium for rigid bodies	[GHSWR] Preface, Chapter 1, Appendix A	Handout 1, 2
2: 8/28	Statics	Concurrent forces: A special case	[GHSWR] Chapter 2	Handout 3, 4
3: 8/31	Statics	Application of statics for systems with moments	[GHSWR] Chapter 3	Handout 5, 6
4: 9/2	Statics	Theory of alternate reference points	Handout	Handout 7, 8
5: 9/4	Statics	Special considerations for two dimensional problems	Handout	Handout 9, 10
6: 9/7		Labor day no class		
7: 9/9	Statics	Equivalent force systems	[GHSWR] Chapter 4	Handout 11, 12
8: 9/11	Statics	Distributed forces	[GHSWR] Chapter 4	Handout 13, 14
9: 9/14	Statics	Applications of statics: Complex systems	[GHSWR] Chapters 5, 6	Handout 15, 16
10: 9/16	Statics	Friction	[GHSWR] Chapters 9	Handout 17, 18
11: 9/18	Axial bar	Introduction to deformable bodies	[G] Preface, Chapter 1	1.8

Lecture	Module	Topic	Reading	HW
12: 9/21	**Exam**	**Midterm 1: Lectures 1-10**		
13: 9/23	Axial bar	1-D stress, strain, equilibrium, constitution	[G] 2.1–2.3	2.3, 2.4
14: 9/25	Axial bar	Axial response	[G] 2.4–2.4.1	2.9, 2.15
15: 9/28	Axial bar	Axial response by direct integration	[G] 2.4.2	2.11, 2.16, 2.31
16: 9/30	Axial bar	Conservation of energy and stress based design	[G] 2.5–2.6	2.33
17: 10/2	Multi-D	General Concepts of Stress	[G] 3.1	3.4
18: 10/5	Multi-D	Pointwise Stress	[G] 3.2	3.8, 3.10, 3.15
19: 10/7	Multi-D	Polar and Spherical Stresses	[G] 3.3	
20: 10/9	Multi-D	General Concepts of Strain	[G] Chapter 4	4.1, 4.5, 4.12
21: 10/12	Multi-D	Generalized Hooke's Law	[G] Chapter 5	5.3, 5.11
22: 10/14	Multi-D	Axial loading as a multi-dimensional phenomena, thin-walled pressure vessels	[G] 6.1	6.4
23: 10/16	Multi-D	Thin walled pressure vessels and St. Venant's Principle	[G] 6.2–6.3	6.7, 6.10, 6.11
24: 10/19	Torsion	Kinematics and Equilibrium of Torsion	[G] 7.1–7.2	7.2
25: 10/21	Torsion	Torsion of Circular Elastic Bars	[G] 7.3–7.4	7.5, 7.8, 7.13, 7.21, 7.26, 7.28
26: 10/23	Torsion	Thin-Walled Torsion	[G] 7.7	7.41, 7.43
27: 10/26	Beams	Kinematics of Bending	[G] 8.1	8.2
28: 10/28	Beams	Equilibrium of Bending	[G] 8.2	8.5
29: 10/30	Beams	Elastic Response of Beams	[G] 8.3	8.8, 8.13, 8.15

Lecture	Module	Topic	Reading	HW
30: 11/2	**Exam**	**Midterm 2: Lectures 10-25**		
31: 11/4	Beams	Beam Deflections by Integration	[G] 8.4	8.19, 8.34
32: 11/6	Beams	Multi-axis Bending	[G] 8.5	8.38
33: 11/9	Beams	Shear Stresses in Beams	[G] 8.6	8.41, 8.47
34: 11/11	Transformations	Transformation of Vectors and Tensors	[G] 9.1–9.2.1	9.1
35: 11/13	Transformations	Principal values, Maximum Shear, Eigenvalues and Eigenvectors	[G] 9.2.2–9.2.3	9.2, 9.4
36: 11/16	Transformations	Mohr's Circle of Stress	[G] 9.2.4–9.2.5	9.6, 9.11
37: 11/18	Transformations	Transformation of Strain	[G] 9.3	9.15, 9.16a, 9.18
38: 11/20	Failure	Yield and Fracture Criteria	[G] 9.4	9.25, 9.26, 9.28, 9.29
39: 11/23	Failure	Stability: Introduction	[G] 12.1–12.2	12.1, 12.4
40: 11/25		Thanksgiving no class		
41: 11/27		Thanksgiving no class		
42: 11/30	Failure	Stability: Introduction	[G] 12.1–12.2	12.5, 12.12
43: 12/2	Failure	Euler Loads for Columns	[G] 12.3	12.14, 12.15
44: 12/4	Review	Review		

Course Staff

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